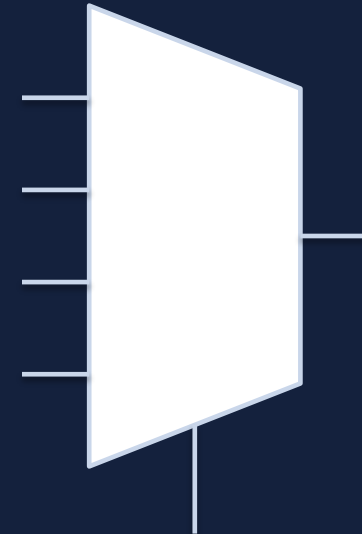


Lecture 16

Review for Exam 2

Combinational analysis and design · decoders and multiplexers · adders, subtractors and overflow · BCD, lookahead and comparators · practice set

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 4 (review)



Today's roadmap

1

§4.4

Analysis, design

Lecture 10

2

§4.9

Decoders, MUXes

Lectures 12–13

3

§4.5

Adders

Lecture 14

4

§4.8

BCD, compare

Lecture 15

By the end of class you can...

- Analyze a circuit and design one from a word problem
- Implement functions with decoders and multiplexers
- Operate the adder-subtractor and detect overflow
- Apply the BCD correction and carry lookahead
- Compare two words with the comparator equations

Agenda for today's session

Intro	Scope	Design	MSI	Break	Arith.	Practice	
0:00	0:05	0:15	0:35	0:55	1:05	1:30	1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homework 7 collected	—
0:05 – 0:15	What the exam covers; common errors	—
0:15 – 0:35	Analysis and design	Reviews 1, 2
0:35 – 0:55	Decoders and multiplexers	Reviews 3, 4
0:55 – 1:05	Break	
1:05 – 1:30	Adders, BCD, lookahead, comparators	Reviews 5–8
1:30 – 1:50	Timed practice set and solutions	Practice set

What Exam 2 covers

Emphasis: combinational blocks

Analysis and design procedure (Lecture 10)

Decoders, encoders, priority encoders (12)

Multiplexers and three-state buffers (13)

Adders, subtractors, overflow, lookahead (14)

BCD adders, multipliers, comparators (15)

Lectures 1–9 underpin every problem: K-maps, forms, NAND and NOR.

Common errors

Wrong select-line order on a MUX: the minterm indexes scramble.

Forgetting that active-low decoder outputs need NAND, not OR.

Reading C_4 as signed overflow; overflow is $C_3 \oplus C_4$.

Adding 6 to a BCD sum of 9 or less.

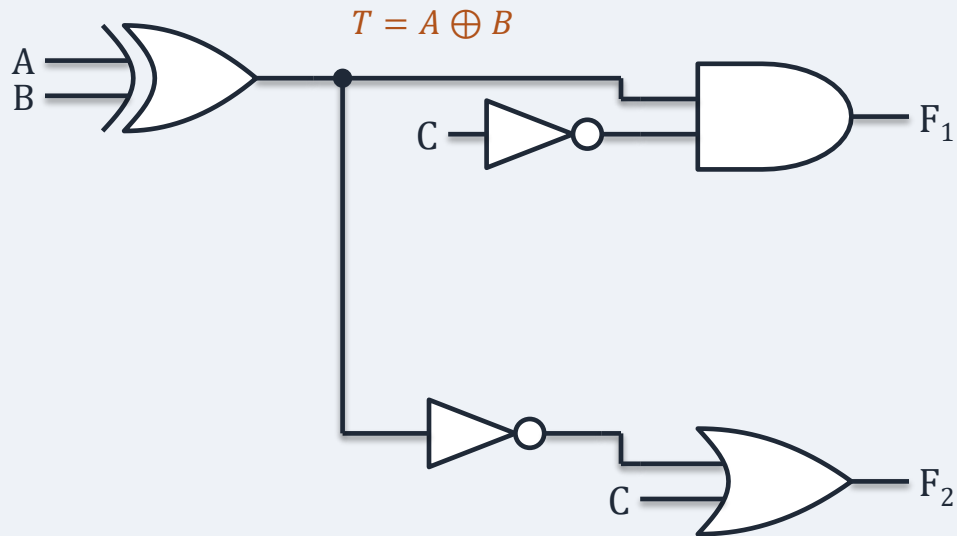
Skipping the labels in circuit analysis.

Review 1 — Analyze a circuit

§4.3

REVIEW

Find F_1 and F_2 as functions of A, B, C and give their minterm lists.



1 Label the XOR output: $T = A \oplus B$

2 Gate for F_1 : $T \text{ AND } C' \rightarrow F_1 = (A \oplus B)C'$

3 $F_1 = 1$ needs $A \neq B$ and $C = 0$: rows 010, 100

4
$$F_1 = \Sigma m(2,4)$$

5 Gate for F_2 : $T' \text{ OR } C \rightarrow F_2 = (A \oplus B)' + C$

6 $F_2 = 1$ when $A = B$ (000, 001, 110, 111) or $C = 1$

7
$$F_2 = \Sigma m(0,1,3,5,6,7)$$

Review 2 — Design a squarer

§4.4

REVIEW

Input $A = A_1A_0$ (0–3). Output the 4-bit square $S_3S_2S_1S_0$. Design the circuit.

A_1A_0	A	A^2	$S_3S_2S_1S_0$
00	0	0	0000
01	1	1	0001
10	2	4	0100
11	3	9	1001

1 Squares of 0–3: 0, 1, 4, 9 = 0000, 0001, 0100, 1001

2 S_3 is 1 only for $A = 3$ (9 = 1001): $S_3 = A_1A_0$

3 S_2 is 1 only for $A = 2$ (4 = 0100): $S_2 = A_1A_0'$

4 S_1 is 0 in every row: $S_1 = 0$

5 S_0 is 1 for $A = 1$ and $A = 3$: $S_0 = A_0$

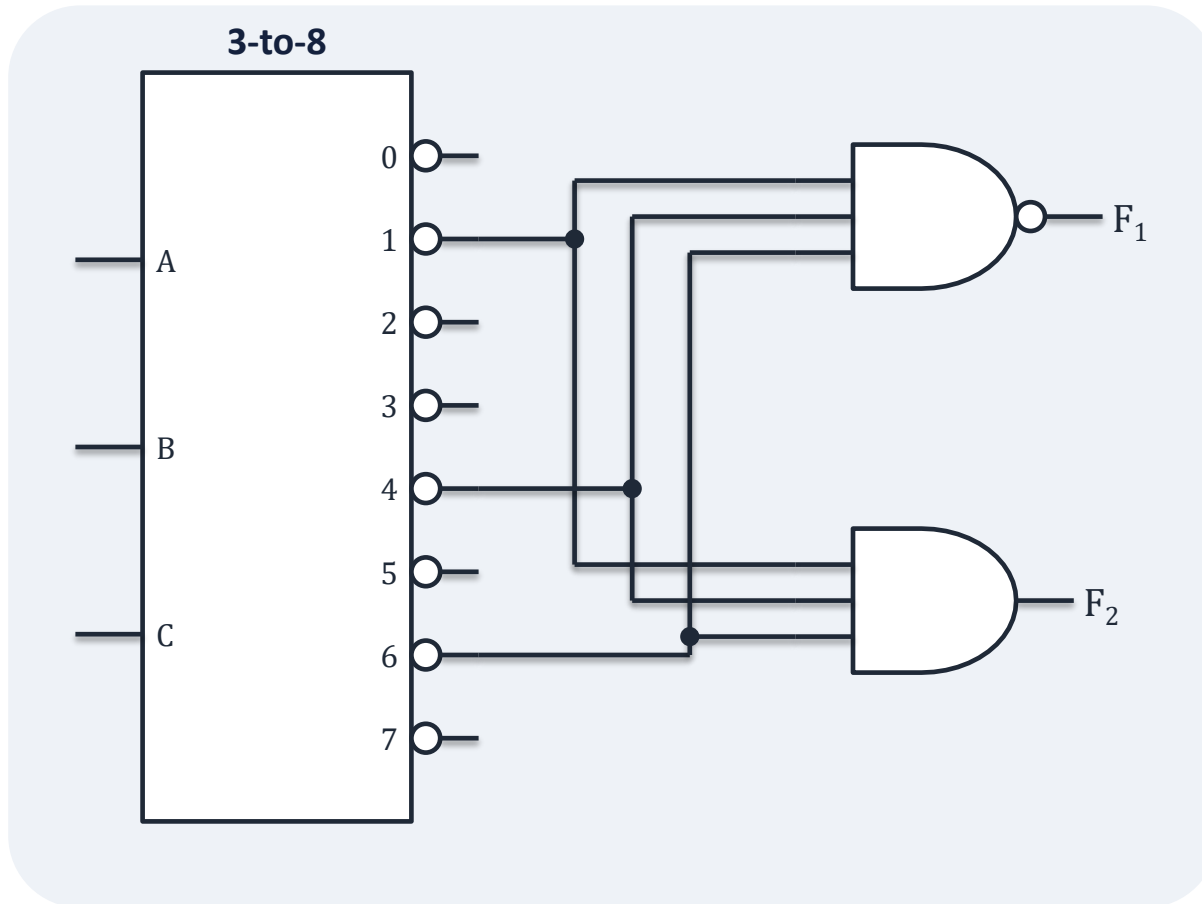
6 Two AND gates and one inverter

Review 3 — Active-low decoder

§4.9

REVIEW

With a 3-to-8 decoder whose outputs are active LOW, build $F_1 = \Sigma m(1,4,6)$ and $F_2 = \Sigma m(0,2,3,5,7)$.



- 1 Active-low: output j is m'_j (0 when selected)
- 2 $F_1 = m_1 + m_4 + m_6 = (m'_1 m'_4 m'_6)'$ (DeMorgan)
- 3 So $F_1 = \mathbf{NAND}$ of outputs 1, 4, 6
- 4 F_2 lists every other minterm: $F_2 = F'_1$
- 5 So $F_2 = m'_1 m'_4 m'_6$: **AND** of outputs 1, 4, 6
- 6 Check 100: output 4 goes low $\rightarrow F_1 = 1, F_2 = 0 \checkmark$
- 7 Two 3-input gates instead of a 5-input one

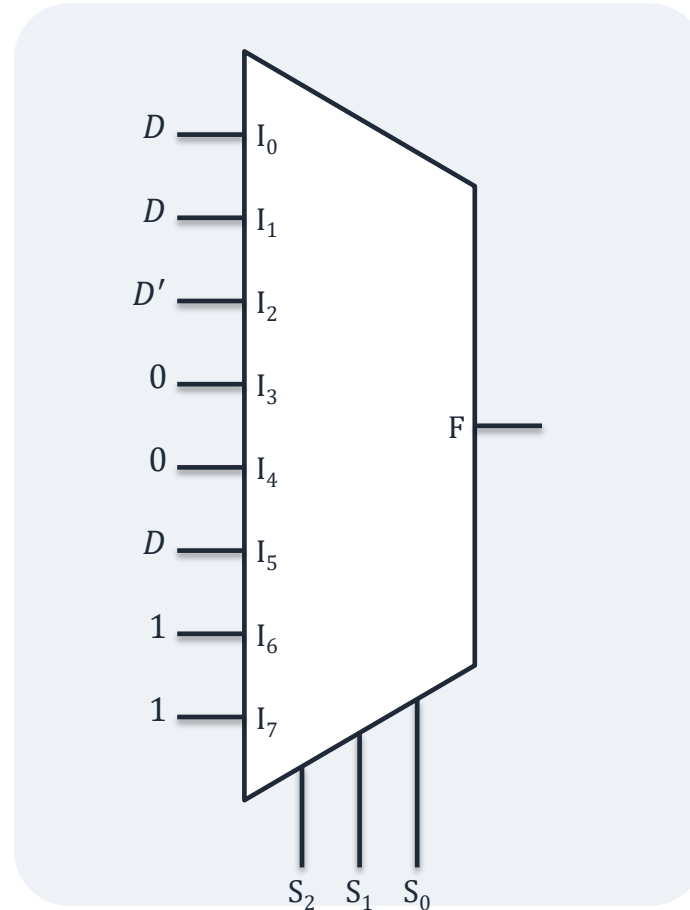
Review 4 — Function on an 8-to-1 MUX

§4.11

REVIEW

Implement $F(A, B, C, D) = \Sigma m(1, 3, 4, 11, 12, 13, 14, 15)$ on an 8-to-1 MUX with A, B, C on the selects.

A B C	D=0	D=1	Input
000	0	1	$I_0 = D$
001	0	1	$I_1 = D$
010	1	0	$I_2 = D'$
011	0	0	$I_3 = 0$
100	0	0	$I_4 = 0$
101	0	1	$I_5 = D$
110	1	1	$I_6 = 1$
111	1	1	$I_7 = 1$



- 1 A, B, C on the selects, D residual: read F at $D = 0, D = 1$
- 2 $000: 0, 1 \rightarrow I_0 = D; 001: 0, 1 \rightarrow I_1 = D$
- 3 $010: 1, 0 \rightarrow I_2 = D'; 011: 0, 0 \rightarrow I_3 = 0$
- 4 $100: 0, 0 \rightarrow I_4 = 0; 101: 0, 1 \rightarrow I_5 = D$
- 5 $110: 1, 1 \rightarrow I_6 = 1; 111: 1, 1 \rightarrow I_7 = 1$
- 6 Check m12: $110 \rightarrow I_6 = 1 \checkmark$

Break

10 minutes — we resume at the 1:05 mark

Next: adders, BCD, lookahead and comparators

Review 5 — Adder-subtractor

§4.5

REVIEW

4-bit adder-subtractor: (a) $M = 1$, $A = 0101$, $B = 0111$. (b) $M = 0$, $A = 0111$, $B = 0010$. Give S , C_4 , V .

(a) $5 - 7$

i	A	$B \oplus M$	C_i	S	C_{i+1}
0	1	0	1	0	1
1	0	0	1	1	0
2	1	0	0	1	0
3	0	1	0	1	0

(b) $7 + 2$

i	A	$B \oplus M$	C_i	S	C_{i+1}
0	1	0	0	1	0
1	1	1	0	0	1
2	1	0	1	0	1
3	0	0	1	1	0

- 1 (a) $M = 1$: $B = 0111 \rightarrow B \oplus 1 = 1000$, $C_0 = 1$
- 2 Add: $0101 + 1000 + 1 = \mathbf{1110}$, $C_4 = 0$
- 3 $C_4 = 0$: a borrow; the result is negative
- 4 1110 : invert 0001 , add $1 \rightarrow 0010 = 2 \rightarrow \mathbf{-2} \checkmark$
- 5 (b) $M = 0$: $0111 + 0010 = \mathbf{1001}$, $C_4 = 0$
- 6 Bits 0–2: $111 + 010$ carries into the sign: $C_3 = 1$
- 7 $V = C_3 \oplus C_4 = 1 \oplus 0 = \mathbf{1}$: overflow
- 8 $+9$ does not fit in $-8 \dots +7$

Review 6 — BCD with a carry in

§4.6

REVIEW

A BCD adder stage receives $A = 6$, $B = 7$ and a carry in of 1. What leaves it?

6	0	1	1	0
+7	0	1	1	1
+carry	0	0	0	1
Z	0	1	1	1
+6	0	1	1	0
out	1	0	1	0

- 1 Binary sum: $0110 + 0111 + 1 = 1110$ (14)
- 2 No carry out of the 4 bits: $K = 0$
- 3 $C = K + Z_8Z_4 + Z_8Z_2 = 0 + 1 + 1 = 1$: correct
- 4 Add 6: $1110 + 0110 = 1\ 0100$
- 5 Ignore that carry; the digit is **0100**
- 6 The decimal carry out is $C = 1$
- 7 $14 = 1 \text{ ten} + 4 \checkmark$

Review 7 — Carry lookahead

§4.5

REVIEW

For $A = 0101$, $B = 0011$, $C_0 = 0$ find P , G , all carries and S .

Bit i	3	2	1	0
A	0	1	0	1
B	0	0	1	1
P	0	1	1	0
G	0	0	0	1
C_i	1	1	1	0
$S = P \oplus C$	1	0	0	0

- 1 $P = A \oplus B$, bits 3–0: 0,1,1,0 $\rightarrow P = 0110$
- 2 $G = AB$, bits 3–0: 0,0,0,1 $\rightarrow G = 0001$
- 3 $C_1 = G_0 + P_0C_0 = 1 + 0 = 1$
- 4 $C_2 = G_1 + P_1C_1 = 0 + 1 \cdot 1 = 1$
- 5 $C_3 = G_2 + P_2C_2 = 0 + 1 \cdot 1 = 1$
- 6 $C_4 = G_3 + P_3C_3 = 0 + 0 \cdot 1 = 0$
- 7 $S = P \oplus C = 0110 \oplus 1110 = \mathbf{1000}$
- 8 $5 + 3 = 8 \checkmark$; bits 1 and 2 propagate the carry

Review 8 — Magnitude comparison

§4.8

REVIEW

Compare $A = 0110$ and $B = 0101$. Which comparator term makes $(A > B) = 1$?

Bit i	3	2	1	0
A	0	1	1	0
B	0	1	0	1
x_i	1	1	0	0

- 1 Bit 3: 0, 0 agree $\rightarrow x_3 = 1$; bit 2: 1, 1 $\rightarrow x_2 = 1$
- 2 Bit 1: $A_1 = 1$, $B_1 = 0$ differ, so $A_1 B'_1 = 1$
- 3 Term $x_3 x_2 A_1 B'_1 = 1 \cdot 1 \cdot 1 \cdot 1 = 1$
- 4 So $(A > B) = 1$
- 5 Bit 0 ($A_0 < B_0$) no longer matters: a higher bit decided
- 6 $6 > 5$ ✓

Practice set — 15 minutes

§4.4

YOUR TURN

≈ 15 min

1. How many AND gates does a 5-bit $\times$ 4-bit array multiplier need?
2. How many outputs does a 5-to-32 decoder have, and how many 3-to-8 decoders build one (with a 2-to-4 for the enables)?
3. BCD: $9 + 8$. Give the decimal carry and the digit.
4. Implement $F(x, y) = (x \oplus y)'$ on a 2-to-1 MUX with x on the select.
5. Why must three-state enables on a bus come from a decoder?

Work alone, as on the exam.

Practice set — Solutions

§4.4

SOLUTION

1. One AND per pair of bits: $5 \times 4 = \mathbf{20}$ ANDs
2. $n = 5$ inputs $\rightarrow 2^5 = \mathbf{32}$ outputs
3. $K - 1 = 3$ adders of 5 bits; product $5 + 4 = 9$ bits
4. $x = 0: F = (0 \oplus y)' = y' \rightarrow I_0 = y'$
5. $x = 1: F = (1 \oplus y)' = y \rightarrow I_1 = y$
6. $0001 + 0110 = 0111$: carry **1**, digit **0111** (17)
7. Two enabled drivers would fight (contention)
8. A decoder enables exactly one output at a time
9. Each 3-to-8 gives 8: $32/8 = \mathbf{4}$ decoders
10. $1001 + 1000 = 1\ 0001$: $K = 1 \rightarrow C = 1$

Exam 2: how to prepare

On the exam

Write ANALYSIS or DESIGN and number your steps.

Label every signal; state which variable is the MSB.

Verify on at least one row before moving on.

Bring

Pencil, eraser, calculator.

Study plan

Redo each Your turn from Lectures 10–15.

Implement one function three ways: gates, decoder, MUX.

Trace the adder-subtractor in both modes with overflow.

Work one BCD sum that needs correction and one that does not.

Review Labs F, G and Hp1–Hp3.

Key takeaways

Analysis

label $\rightarrow$ substitute $\rightarrow$ table

Decoder

active-low $\rightarrow$ NAND

MUX

inputs 0, 1, r , r'

Subtract

$$A + B' + 1; V = C_3 \oplus C_4$$

BCD

$$+6 \text{ when } C = K + Z_8Z_4 + Z_8Z_2$$

Compare

first differing bit decides

Every block is a truth table you can build three ways.

Homework #8 due at the start of Lecture 19

This afternoon's lab: Lab Hp1 — Calculator 1: half and full adders

Next: Exam 2, then latches and sequential circuits